

Requirements Table: Academic Elective Bidding & Allocation System

Functional Requirements (FR-001 to FR-005)

Requirement ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow students to select up to 5 ranked elective courses and distribute exactly 100 bidding credits across them during an active bidding window.	High	Pass: Total credit sum equals 100 and preferences are ranked sequentially. Fail: Total credit sum does not equal 100, or invalid course selections are submitted.	Enables core market-based dynamic course registration driven by student preferences.

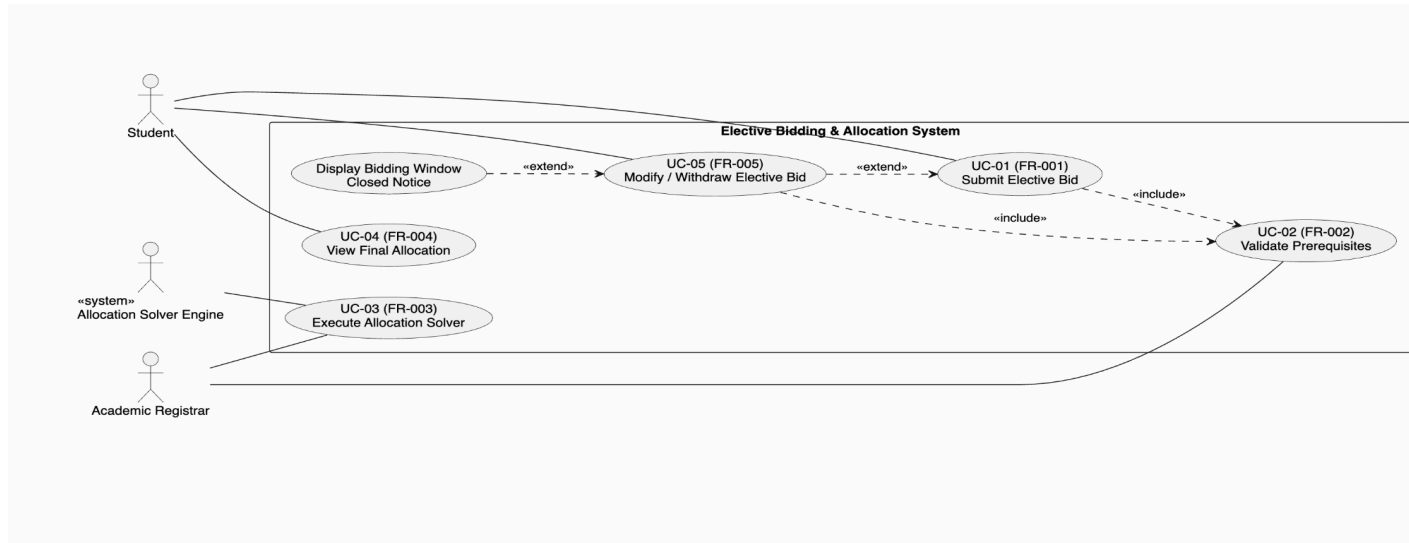
FR-002	Functional	The system shall validate student prerequisite course completions before allowing a credit bid to be finalized for any elective.	High	Pass: Bid accepted if all prerequisite courses are marked complete in academic records. Fail: Bid blocked if any required prerequisite is incomplete.	Prevents academic progression issues by ensuring students possess required foundational knowledge.
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FR-003	Functional	The system shall execute an automated allocation engine that resolves seat capacities and timetable conflicts to assign courses based on credit bids.	High	Pass: Seat allocation runs, resolving all schedule collisions without exceeding capacity limits. Fail: Overbooking occurs or schedule overlaps remain unresolved.	Automates fair, collision-free course assignment across the department without manual intervention.
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FR-004	Functional	The system shall allow students and the Academic Registrar to view finalized elective allocation results and schedule summaries.	Medium	Pass: Student can access a read-only list of allocated courses and timetables. Fail: Results are hidden, misaligned, or accessible prior to official release.	Provides transparency for students to review their confirmed course schedules and for admins to audit outcomes.
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NFR-001	Performance	The elective allocation solver shall process up to 5,000 student bids and resolve schedule conflicts in under 30 seconds under peak load.	High	Pass: Automated solver completes full execution within 30 seconds for 5,000 concurrent bids. Fail: Processing time exceeds 30 seconds or execution times out.	Ensures system scalability and fast execution during high-concurrency end-of-cycle allocation runs.
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NFR-002	Security & Access Control	The system shall enforce role-based authentication to ensure students can only view/modify their own bids, while preventing post-deadline modifications.	High	Pass: Unauthorized access attempts fail; bidding form becomes read-only precisely at the deadline. Fail: User accesses another student's bid or modifies a bid after window closure.	Secures student data integrity and guarantees fairness by enforcing strict bidding timeframe rules.
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Use-Case Flow Specification

Academic Elective Bidding & Allocation System
 Lab 1: Requirements Engineering & UML Use-Case Modelling

Field: Use Case

Details: Modify / Withdraw Elective Bid (UC-05 / FR-005)

Field: Primary Actor

Details: Student

Field: Secondary Actor

Details: None

Field: Includes

Details: Validate Prerequisites «include» — re-run whenever the bid is modified (not required for a withdrawal)

Field: Trigger

Details: Student decides to change or cancel a previously submitted elective bid before the bidding window closes.

Field: Preconditions

1. The student has already submitted an elective bid for the current bidding cycle (Use Case: Submit Elective Bid is complete).
2. The bidding window is still open (current time is before the published deadline).

3. The student is authenticated and the bid being modified belongs to them.

Field: Postconditions

Success: The student's bid record reflects the new credit distribution and ranked preferences (or is marked withdrawn), fully replacing the previous version. The system confirms the change and timestamps it.

Failure: No change is applied. The student's original bid remains exactly as it was, and the system informs the student why the update was rejected.

Main Success Scenario

1. Student logs in and navigates to "My Bids."
2. System displays the student's current bid summary (credit distribution and ranked elective preferences).
3. Student selects "Modify Bid" or "Withdraw Bid."
4. If modifying, the student re-enters the new credit distribution and/or reorders ranked preferences.
5. System validates that the new total does not exceed 100 credits and re-checks prerequisite eligibility («include»: Validate Prerequisites).
6. System saves the updated bid, overwriting the previous submission, and records a timestamp for the change.
7. System displays a confirmation message with the updated (or withdrawn) bid summary.
8. Use case ends successfully.

Alternate Flow

5a: Bidding Window Has Closed

5a1. At step 5, the system detects that the current time is past the published bidding deadline.

5a2. System rejects the modification/withdrawal request and displays a message explaining that the window has closed.

5a3. The student's original bid remains unchanged.

5a4. Student is redirected to a read-only view of their existing bid. Use case ends without applying any change.